

Ex. No.: 10A

FIFO PAGE REPLACEMENT

Aim:

To find out the number of page faults that occur using First-in First-out (FIFO) page replacement technique.

Algorithm:

1. Declare the size with respect to page length
2. Check the need of replacement from the page to memory
3. Check the need of replacement from old page to new page in memory
4. Form a queue to hold all pages
5. Insert the page require memory into the queue
6. Check for bad replacement and page fault
7. Get the number of processes to be inserted
8. Display the values

Program:

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Q1

FIFO Page Replacement Algorithm

Completed

Write a C program to implement the FIFO Page Replacement Algorithm. The program should take the number of frames and a reference string (sequence of page requests) as input. For each page in the reference string, determine if it results in a 'Page Hit' or a 'Page Fault'. Display the state of the frames after each request and calculate the total number of page faults and hits.

Input Format

1. The first line of input contains an integer representing the number of pages in the reference string.
2. The second line contains the page reference string as a sequence of integers.
3. The third line contains an integer representing the number of available frames.

Output Format

The total number of Page Faults and Page Hits are displayed

Sample Input 1

Select Language: C (GCC 9.2.0)

```
1 #include <stdio.h>
2
3 int main() {
4     int n, frames, i, j, k, pos = 0;
5
6     printf("Enter number of pages in reference string: \n");
7     scanf("%d", &n);
8
9     int ref[n];
10    printf("Enter the reference string: \n");
11    for(i = 0; i < n; i++) {
12        scanf("%d", &ref[i]);
13    }
14
15    printf("Enter number of frames: \n");
16    scanf("%d", &frames);
17
18    int frame[frames];
19
20    // Initialize frames with -1
21    for(i = 0; i < frames; i++) {
22        frame[i] = -1;
23    }
24}
```

Test Cases 2

Run Sample Cases

Run All Cases

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Completed

FIFO Page Replacement Algorithm

Write a C program to implement the FIFO Page Replacement Algorithm. The program should take the number of frames and a reference string (sequence of page requests) as input. For each page in the reference string, determine if it results in a 'Page Hit' or a 'Page Fault'. Display the state of the frames after each request and calculate the total number of page faults and hits.

Input Format

1. The first line of input contains an integer representing the number of pages in the reference string.
2. The second line contains the page reference string as a sequence of integers.
3. The third line contains an integer representing the number of available frames.

Output Format

The total number of Page Faults and Page Hits are displayed

Sample Input 1

Select Language: C (GCC 9.2.0)

```
29
30 // Check if page already in frame (HIT)
31 for(j = 0; j < frames; j++) {
32     if(frame[j] == ref[i]) {
33         found = 1;
34         hits++;
35         break;
36     }
37 }
38
39 // If not found -> FAULT
40 if(!found) {
41     frame[pos] = ref[i];
42     pos = (pos + 1) % frames; // FIFO rotation
43     faults++;
44 }
45
46 // Output
47 printf("\nTotal Page Faults: %d", faults);
48 printf("\nTotal Page Hits: %d", hits);
49
50
51 return 0;
52 }
```

Test Cases 2

Run Sample Cases

Run All Cases

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Completed

FIFO Page Replacement Algorithm

Write a C program to implement the FIFO Page Replacement Algorithm. The program should take the number of frames and a reference string (sequence of page requests) as input. For each page in the reference string, determine if it results in a 'Page Hit' or a 'Page Fault'. Display the state of the frames after each request and calculate the total number of page faults and hits.

Input Format

1. The first line of input contains an integer representing the number of pages in the reference string.
2. The second line contains the page reference string as a sequence of integers.
3. The third line contains an integer representing the number of available frames.

Output Format

The total number of Page Faults and Page Hits are displayed

Sample Input 1

Test Cases 2

Run Sample Cases

Run All Cases

Custom Test Case

> Custom Test

Sample Test Case

Test Case - 1

Input :

7
1 3 0 3 5 6 3
3

Expected Output :

Enter number of pages in reference string:
Enter the reference string:
Enter number of frames:

Total Page Faults: 6
Total Page Hits: 1

Output :

Test Case - 2

Result:

Thus the program to find the number of page faults that occur using FIFO replacement technique has been successfully executed.

Ex. No.: 10B

LRU PAGE REPLACEMENT

Aim:

LRU To write a c program to implement LRU page replacement algorithm.

Algorithm:

- 1: Start the process
- 2: Declare the size
- 3: Get the number of pages to be inserted
- 4: Get the value
- 5: Declare counter and stack
- 6: Select the least recently used page by counter value
- 7: Stack them according to the selection.
- 8: Display the values
- 9: Stop the process

Program:

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Q1

FIFO Page Replacement Algorithm

Completed

Write a C program to implement the FIFO Page Replacement Algorithm. The program should take the number of frames and a reference string (sequence of page requests) as input. For each page in the reference string, determine if it results in a 'Page Hit' or a 'Page Fault'. Display the state of the frames after each request and calculate the total number of page faults and hits.

Input Format

1. The first line of input contains an integer representing the number of pages in the reference string.
2. The second line contains the page reference string as a sequence of integers.
3. The third line contains an integer representing the number of available frames.

Output Format

The total number of Page Faults and Page Hits are displayed

Sample Input 1

Select Language: C (GCC 9.2.0)

```
1 #include <stdio.h>
2
3 int main() {
4     int n, frames, i, j, k, pos = 0;
5
6     printf("Enter number of pages in reference string: \n");
7     scanf("%d", &n);
8
9     int ref[n];
10    printf("Enter the reference string: \n");
11    for(i = 0; i < n; i++) {
12        scanf("%d", &ref[i]);
13    }
14
15    printf("Enter number of frames: \n");
16    scanf("%d", &frames);
17
18    int frame[frames];
19
20    // Initialize frames with -1
21    for(i = 0; i < frames; i++) {
22        frame[i] = -1;
23    }
24
```

Test Cases 2

Run Sample Cases

Run All Cases

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FIFO Page Replacement Algorithm

Write a C program to implement the FIFO Page Replacement Algorithm. The program should take the number of frames and a reference string (sequence of page requests) as input. For each page in the reference string, determine if it results in a 'Page Hit' or a 'Page Fault'. Display the state of the frames after each request and calculate the total number of page faults and hits.

Input Format

- The first line of input contains an integer representing the number of pages in the reference string.
- The second line contains the page reference string as a sequence of integers.
- The third line contains an integer representing the number of available frames.

Output Format

The total number of Page Faults and Page Hits are displayed

Sample Input 1

Q1

Save

Select Language: C (GCC 9.2.0)

```

29
30 // Check if page already in frame (HIT)
31 for(j = 0; j < frames; j++) {
32     if(frame[j] == ref[i]) {
33         found = 1;
34         hits++;
35         break;
36     }
37 }
38
39 // If not found -> FAULT
40 if(!found) {
41     frame[pos] = ref[i];
42     pos = (pos + 1) % frames; // FIFO rotation
43     faults++;
44 }
45
46 // Output
47 printf("\nTotal Page Faults: %d", faults);
48 printf("\nTotal Page Hits: %d", hits);
49
50
51 return 0;
52

```

Test Cases 2

Run Sample Cases

Run All Cases

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Completed

FIFO Page Replacement Algorithm

Write a C program to implement the FIFO Page Replacement Algorithm. The program should take the number of frames and a reference string (sequence of page requests) as input. For each page in the reference string, determine if it results in a 'Page Hit' or a 'Page Fault'. Display the state of the frames after each request and calculate the total number of page faults and hits.

Input Format

- The first line of input contains an integer representing the number of pages in the reference string.
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- The third line contains an integer representing the number of available frames.

Output Format

The total number of Page Faults and Page Hits are displayed

Sample Input 1

Q1

Save

Test Cases 2

Run Sample Cases

Run All Cases

Custom Test Case

> Custom Test

Sample Test Case

> Test Case - 1

Input :

7

1 3 0 3 5 6 3

3

Expected Output :

Enter number of pages in reference string:

Enter the reference string:

Enter number of frames:

Total Page Faults: 6

Total Page Hits: 1

Output :

> Test Case - 2

Result:

Thus the program to implement LRU page replacement algorithm has been successfully executed.